Proposed x-sensitive Miller-primitive half-root screen

2026-10-27 | Read-only source orientation and hand derivations | No experiment executed

Public primitive

Binary D recurrence

$Z(R) = -D_r(R) / r$

Existing zeta/cocycle prior art

O(log r) charged setup/evaluation

Fixed public decoder

$c = Z(P) / y(P)$

$g = Z + cy[\alpha + \beta c(x - x(P))]$

$H = (1 - \chi(g(R)/g(P))) / 2$

40 fixed α, β templates

Mandatory fixed fixture

E/F11: $y^2 = x^3 + x + 6$, r = 13

12 generators × 13 targets × 40

6,240 planned cases before controls

Undefined values remain failures

Derived local obstruction: do not repeat the one-variable family

On the mandatory F11 fixture, 5P=(3,6), 6P=(7,9), Z(5P)=4 and Z(6P)=6. Both have Z/y=8, but $\chi(6/9)=\chi(8)=-1$. Therefore every normalized-character decoder $g=y\cdot F(Z/y)$ gives opposite signs at these two lower-half points when defined. A zero or pole is undefined. The new x term escapes this particular fiber collision; its ordering is untested.

Cost and acceptance remain separate

Each predicate call costs O(log r + log p) group/field operations, with inversions and χ powering charged. Conditional scalar recovery needs O(log r) calls plus target doubling and final verification. Validate full cocycle, oddness, all coordinate scalings, generator reindexing and the exact two-pulse law, including O and boundary exceptions. Truth-derived halves are validator-only controls.

Scope and sources

A finite pass advances only that fixed circuit to an ordering-proof question. Failure resolves its stated template, not all coordinate circuits. No new cohort, statistical fit, scalar-table advice, performance qualification, novelty verdict or ledger status transition is supplied. Complete formulas and source hashes: proposal.md, proposal.json and source-manifest.json.

Prior: IDEA-20261016-09fd08; existing ordered-root audit; Voloch, Acta Arith. 79 (1997), 1–6.

Primary author PDF read: https://www.math.canterbury.ac.nz/~f.voloch/Pdfs/zeta3.pdf